

# Users, Roles & Permissions Guide

Nyongeza ya Database Schema — nani anaweza kufanya nini, jinsi logs zinavyotengenezwa, na jinsi support inavyoshughulikia matatizo ya watumiaji.

Hii document inajenga juu ya **database\_schema.pdf** iliyotangulia. Inaeleza ngazi nne za watumiaji (ADMIN, RESERVATION\_MANAGER, SALES\_PERSON, GUIDE), mipaka ya kila mmoja, jinsi kila kitendo kinavyotunzwa kwenye **audit\_logs** na **notifications**, 'cheat sheet' ya vitendo vyote vya system, na mtiririko wa kumsaidia mtumiaji (support flow) ikiwemo password reset kwa kutumia column mpya **password\_reset\_token** na **password\_expire**.

## Muhtasari wa Roles Nne

| Role                | Kusudi                                                                                                            | Mfano wa Mtu                                      |
|---------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| ADMIN               | Msimamizi mkuu wa kampuni (tenant) — anaona na kusimamia karibu kila kitu ndani ya company yake.                  | Mmiliki wa kampuni ya safari / Ops Manager        |
| RESERVATION_MANAGER | Anashughulikia uhusiano na properties/lodges na kuthibitisha malazi (bookings/invoices).                          | Reservations desk                                 |
| SALES_PERSON        | Anahudumia client moja kwa moja — kuanzia mawasiliano ya kwanza hadi malipo.                                      | Travel consultant / Sales agent                   |
| GUIDE               | Anapokea taarifa za safari aliyopangiwa tu, kwa ajili ya kuiongoza uwanjani.                                      | Field guide / Driver-guide                        |
| PLATFORM ADMIN      | Anasimamia PLATFORM NZIMA (makampuni yote) — nje kabisa ya companies/users. Ni ngazi ya juu kuliko Company Admin. | Wewe (mmiliki wa software) / timu yako ya support |

## Permissions Kwa Kila Role

### ADMIN

| ✓ ANAWEZA KUFANYA                                                                       | ✗ HAWEZI KUFANYA                                                                        |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Kuongeza/kuedit/kuzima (deactivate) users na kuwapa roles                               | Kuona data za company (tenant) NYINGINE — tenant isolation ni ngumu (hard boundary)     |
| Kuona clients, safaris, bookings, invoices, payments za sales persons WOTE wa kampuni   | Kuunda/kubadilisha bei za subscription_plans zenyewe — hiyo ni ya Platform/SaaS owner   |
| Kusimamia properties: kuongeza, ku-verify, kuedit categories/price tiers/amenities/tags | Kufuta rekodi za audit_logs (ni append-only kwa uadilifu wa historia)                   |
| Kuona na kutatua accommodation_inquiries zote                                           | Kuzima 2FA/ security policy za platform nzima                                           |
| Kuona audit_logs zote za kampuni (read-only, kwa uchunguzi)                             | Kuedit embedding model / AI model configuration ya msingi bila ruhusa ya platform-level |
| Kuchagua/kubadili company_subscription (upgrade/downgrade kati ya plans zilizopo)       |                                                                                         |
| Kuidhinisha/kufuta agent_wisdom kabla haijachapishwa kwa wengine                        |                                                                                         |
| Kupokea notifications muhimu za kampuni nzima                                           |                                                                                         |
| Ku-trigger password reset kwa niaba ya user, au kufungua account iliyofungwa            |                                                                                         |
| Kupakia knowledge_documents kwa ajili ya RAG/AI                                         |                                                                                         |

### RESERVATION\_MANAGER

| ✓ ANAWEZA KUFANYA                                               | ✗ HAWEZI KUFANYA                                                                        |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Kuona accommodation_requirements zilizoletwa na sales persons   | Kuongeza/kuedit/kufuta users au kubadilisha roles                                       |
| Kuunda/kuedit accommodation_bookings na kubadilisha status yake | Kuedit taarifa za msingi za client (jina, mawasiliano) — anaona tu kwa ajili ya booking |
| Kuhamisha status: PROVISIONAL → INVOICED → PAID → CONFIRMED     | Kurekodi payments moja kwa moja (ni kazi ya Sales Person)                               |

| ✓ ANAWEZA KUFANYA                                                                    | ✗ HAWEZI KUFANYA                                                   |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Kuunda invoices kwa ajili ya bookings                                                | Kubadilisha company_subscription au bei za plans                   |
| Kujibu accommodation_inquiries kutoka kwa sales persons                              | Kuona audit_logs za sehemu nyingine zisizohusu properties/bookings |
| Kuongeza property mpya, kuedit amenities/tags/peer_relationships, ku-verify property | Kufuta invoice/booking iliyokwisha CONFIRMED bila idhini ya Admin  |
| Kuona notifications: invoice expiring, missing night, alternative suggested          |                                                                    |
| Kuandika agent_wisdom kuhusu properties                                              |                                                                    |

## SALES\_PERSON

| ✓ ANAWEZA KUFANYA                                                         | ✗ HAWEZI KUFANYA                                                              |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Kuongeza/kuedit clients WAKE mwenyewe, guests, guest_requirements         | Kuona/kuedit clients wa sales person MWINGINE (isipokuwa Admin amemruhusu)    |
| Kuongeza special_occasions na flight_details za client wake               | Kuongeza property mpya au kubadilisha peer_relationships                      |
| Kuunda safaris na itinerary_days kwa client wake                          | Kuthibitisha (CONFIRM) accommodation_booking — hiyo ni ya Reservation Manager |
| Kuunda accommodation_requirements (kuomba malazi kwa Reservation Manager) | Kuona/kuedit users wengine au roles                                           |
| Kutuma accommodation_inquiries kwa Reservation Manager                    | Kufuta payment/invoice iliyokwisha rekodiwa bila idhini ya Admin              |
| Kurekodi payments za client wake na kuzi-allocate kwenye invoices husika  | Kuona audit_logs za company                                                   |
| Kuona status ya bookings/invoices za client wake (real-time)              |                                                                               |
| Kuandika notes na agent_wisdom kutokana na uzoefu wake                    |                                                                               |

## GUIDE

| ✓ ANAWEZA KUFANYA                                                             | ✗ HAWEZI KUFANYA                                                |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Kuona guide_brief ya safari aliyopangiwa (client overview, itinerary, malazi) | Kuona taarifa za fedha (invoices, payments, pricing) kabisa     |
| Kuona guest_requirements muhimu (dietary, medical, mobility) za safari yake   | Kuona clients/safaris ambazo hajapangiwa                        |
| Kuona emergency contacts na vouchers zinazohusiana na brief yake              | Kuedit chochote kwenye client, safari, booking au property data |
| Kuthibitisha kupokea brief (kuweka status/received)                           | Kuona dashboard/reports za admin au audit_logs                  |
|                                                                               | Kuwasiliana moja kwa moja na properties kupitia system          |

## PLATFORM ADMIN

Tofauti na roles nne za juu (ambazo zote ziko ndani ya *users* table na zina *company\_id*), Platform Admin yuko kwenye table yake mwenyewe (*platform\_admins*), NJE ya companies zote. Ndiyo ngazi ya juu kabisa ya usimamizi wa software nzima.

| ✓ ANAWEZA KUFANYA                                                                                        | ✗ HAWEZI KUFANYA                                                                                                       |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Kuongeza kampuni mpya (companies) kwenye platform na kuunda Company Admin wake wa kwanza                 | Kuona/kuedit operational data ya ndani ya kampuni (clients, guests, safaris) bila sababu ya kiufundi/support na ridhaa |
| Kuzima/kusimamisha (suspend) kampuni isiyolipa au inayovunja masharti ya matumizi                        | Kuidhinisha properties au kuandika agent_wisdom kwa niaba ya kampuni                                                   |
| Kuunda/kuedit subscription_plans (bei, features, max_users) za platform nzima                            | Kufuta rekodi za audit_logs (bado ni append-only, hata kwa Platform Admin)                                             |
| Kuona/kufuatilia company_subscriptions, subscription_invoices na subscription_payments za MAKAMPUNI YOTE | Kuingilia maamuzi ya kibiashara ya kampuni binafsi (bei zao za safari, wateja wao)                                     |
| Kurekebisha auto_renew au default_payment_method ya kampuni (kwa ombi lao, kwa ajili ya msaada)          | Kufanya kazi za kila siku za Sales/Reservation/Guide (si role ya operations)                                           |
| Kuona audit_logs za matukio ya platform-level (kuundwa/kuzimwa kwa kampuni, mabadiliko ya plans)         |                                                                                                                        |
| Kutatua support_tickets za category=TECHNICAL/BILLING zinazovuka makampuni                               |                                                                                                                        |
| Kuangalia system health na analytics ya jumla (idadi ya makampuni, mapato ya jumla ya SaaS, n.k.)        |                                                                                                                        |
| Ku-reset password ya Company Admin kama njia nyingine za msaada wa ndani ya kampuni zimeshindikana       |                                                                                                                        |

# Jinsi Logs Zinavyotengenezwa

Mfumo unatumia **mbinu mbili zinazokamilishana** kutunza historia:

## 1. audit\_logs (rekodi ya kina, kwa kila kitendo kinachobadilisha data)

Kila UPDATE/CREATE/DELETE kwenye tables nyeti (users, clients, safaris, accommodation\_bookings, invoices, payments, properties, subscriptions, n.k.) inatengeneza record moja kwenye audit\_logs yenye: *user\_id* aliyefanya kitendo, *action* (mf. UPDATE\_PAYMENT), *entity\_type/entity\_id*, *old\_values* na *new\_values* (JSONB — kabla na baada), *ip\_address*, *user\_agent*, na *created\_at*. audit\_logs ni **append-only** — hairuhusiwi kufutwa au kuedit, hata na Admin, ili kudumisha uadilifu wa historia.

## 2. Column ya version (kwenye tables muhimu)

Tables kama payments, invoices, accommodation\_bookings, properties, price\_tiers, subscription\_plans n.k. zina column ya *version* inayoongezeka kila record ikibadilishwa. Hii inatoa namba rahisi ya "kizazi" cha record, wakati audit\_logs inatoa maelezo kamili ya kilichobadilika kwenye kila version hiyo.

## 3. notifications (matukio yanayohitaji hatua/taarifa)

Tofauti na audit\_logs (ambayo ni ya kihistoria/ukaguzi), notifications zinatengenezwa na "business rules" / scheduled jobs kuashiria kitu kinachohitaji hatua — mf. invoice inakaribia *due\_date*, "missing night" kwenye itinerary, payment gap, request iliyokaa >48h bila majibu, au property mbadala imependekezwa. Notification inatumwa kwa *user\_id* husika (mf. Sales Person au Reservation Manager) yenye priority na inaweza kusomwa (*read\_at*).

## 4. Password reset (mfano halisi wa matumizi ya token + logs)

(a) User anabofya "Nimesahau Password" → (b) System inatengeneza *password\_reset\_token* ya kipekee (random, imesimbwa/hashed) na kuweka *password\_expire* (mfano dakika 30–60 kuanzia sasa) → (c) Email/SMS yenye link inatumwa kwa user → (d) User anaweka password mpya kabla ya *password\_expire* → (e) Token inafutwa mara moja (single-use) na *password\_hash* mpya inahifadhiwa → (f) audit\_logs inarekodi PASSWORD\_RESET action. Token ikiisha muda (expired) au ikitumika mara moja, haitatumika tena — user atalazimika kuomba mpya.

## Cheat Sheet — Vitendo Vyote vya System

Jedwali hili linaonyesha kila kitendo kikuu cha system, ni role gani inaruhusiwa kukifanya, kama kinaingia kwenye audit\_logs, na kama kinasababisha notification.

| Kitendo (Action)                               | Nani Anaweza                | Audit Log?            | Notification?                           |
|------------------------------------------------|-----------------------------|-----------------------|-----------------------------------------|
| Kuunda user mpya                               | ADMIN                       | Ndiyo                 | Hapana                                  |
| Kubadilisha role ya user                       | ADMIN                       | Ndiyo                 | Anaarifiwa user husika                  |
| Kuzima (deactivate) user                       | ADMIN                       | Ndiyo                 | Anaarifiwa user husika                  |
| Kuomba password reset                          | Yeyote (self-service)       | Ndiyo (token created) | Email/SMS ya token                      |
| Kuweka password mpya kwa token                 | Yeyote mwenye token halali  | Ndiyo                 | Email ya uthibitisho                    |
| Kuongeza client mpya                           | SALES_PERSON, ADMIN         | Ndiyo                 | Hapana                                  |
| Kuongeza guest / guest_requirement             | SALES_PERSON, ADMIN         | Ndiyo                 | Hapana                                  |
| Kuunda safari + itinerary_days                 | SALES_PERSON, ADMIN         | Ndiyo                 | Hapana                                  |
| Kuunda accommodation_requirement               | SALES_PERSON, ADMIN         | Ndiyo                 | Anaarifiwa Reservation Manager          |
| Kuunda accommodation_booking (PROVISIONAL)     | RESERVATION_MANAGER, ADMIN  | Ndiyo                 | Anaarifiwa Sales Person                 |
| Kubadilisha status booking → INVOICED          | RESERVATION_MANAGER, ADMIN  | Ndiyo                 | Invoice inatengenezwa + arifa           |
| Kubadilisha status booking → CONFIRMED         | RESERVATION_MANAGER, ADMIN  | Ndiyo                 | Anaarifiwa Sales Person                 |
| Kurekodi payment                               | SALES_PERSON, ADMIN         | Ndiyo                 | Arifa ikiwa invoice imelipwa kikamilifu |
| Kugawanya payment kwenye invoices (allocation) | SALES_PERSON, ADMIN, System | Ndiyo                 | Hapana                                  |
| Invoice inakaribia due_date (auto)             | System (scheduled job)      | N/A                   | Sales + Reservation Manager             |
| 'Missing night' kwenye itinerary (auto-detect) | System (scheduled job)      | N/A                   | Reservation Manager                     |
| Kutuma accommodation_inquiry                   | SALES_PERSON                | Ndiyo                 | Reservation Manager                     |
| Kujibu accommodation_inquiry                   | RESERVATION_MANAGER         | Ndiyo                 | Sales Person                            |
| Kuongeza property mpya                         | ADMIN, RESERVATION_MANAGER  | Ndiyo                 | Hapana                                  |
| Ku-verify property                             | ADMIN, RESERVATION_MANAGER  | Ndiyo                 | Hapana                                  |
| Kuongeza tag / peer_relationship               | ADMIN, RESERVATION_MANAGER  | Ndiyo                 | Hapana                                  |

| Kitendo (Action)                                  | Nani Anaweza                                       | Audit Log?             | Notification?                      |
|---------------------------------------------------|----------------------------------------------------|------------------------|------------------------------------|
| Kuandika agent_wisdom                             | SALES_PERSON, RESERVATION_MANAGER, ADMIN           | Ndiyo                  | Hapana                             |
| Kupakia knowledge_document                        | ADMIN                                              | Ndiyo                  | Hapana                             |
| Kuuliza AI swali (ai_interaction)                 | Wote (kulingana na scope ya role)                  | Ndiyo (ni log yenyewe) | Hapana                             |
| Kutoa AI feedback (👍/👎)                           | Wote                                               | Ndiyo                  | Hapana                             |
| Kutengeneza na kutuma guide_brief                 | RESERVATION_MANAGER, ADMIN                         | Ndiyo                  | Guide anaarifiwa                   |
| Guide kuthibitisha kupokea brief                  | GUIDE                                              | Ndiyo                  | Reservation Manager (optional)     |
| Kubadilisha company_subscription                  | ADMIN                                              | Ndiyo                  | Admin + (baadaye) billing team     |
| Kuangalia audit_logs                              | ADMIN (read-only)                                  | Hapana (ni soma tu)    | Hapana                             |
| Kufungua support_ticket (mf. login/payment issue) | Yeyote, au Admin kwa niaba                         | Ndiyo                  | Anaarifiwa assigned_to             |
| Kufunga (RESOLVE) support_ticket                  | ADMIN, RESERVATION_MANAGER (kulingana na category) | Ndiyo                  | Anaarifiwa raised_by               |
| Kuongeza kampuni mpya kwenye platform             | PLATFORM ADMIN                                     | Ndiyo                  | Company Admin wa kwanza anaarifiwa |
| Kusimamisha (suspend) kampuni                     | PLATFORM ADMIN                                     | Ndiyo                  | Company Admin wote wanaarifiwa     |
| Kuunda/kuedit subscription_plan                   | PLATFORM ADMIN                                     | Ndiyo                  | Hapana                             |
| Kuunda subscription_invoice kwa kampuni           | PLATFORM ADMIN, System (scheduled)                 | Ndiyo                  | Company Admin anaarifiwa           |
| Kurekodi/kuthibitisha subscription_payment        | PLATFORM ADMIN, System (gateway webhook)           | Ndiyo                  | Company Admin anaarifiwa           |
| Kutatua support_ticket ya TECHNICAL/BILLING       | PLATFORM ADMIN                                     | Ndiyo                  | Anaarifiwa raised_by               |

# Support Flow — Jinsi Mtumiaji Anavyosaidiwa

## Ngazi ya 1 — Self-service (papo hapo, bila kuhitaji mtu)

- Password iliyosahaulika → "Forgot Password" → password\_reset\_token + password\_expire → email/SMS → user anaweka password mpya mwenyewe.
- Account "locked" kwa sababu ya majaribio mengi ya kuingia → system inaweza kufungua yenyewe baada ya muda fulani (mfano dakika 15) au kupitia reset ya password.

## Ngazi ya 2 — Msaada wa ndani ya timu (Sales ↔ Reservation Manager)

- Tatizo la uendeshaji (mf. booking haijaonekana, invoice haijatengenezwa) → linatatuliwa kupitia *accommodation\_inquiries* moja kwa moja kati ya Sales Person na Reservation Manager, likiwa na *subject*, *message* na *priority*. Status inafuatiliwa: OPEN → IN\_PROGRESS → RESOLVED → CLOSED.

## Ngazi ya 3 — Escalation kwa Admin wa kampuni

- Ikiwa Ngazi ya 2 haijatatua tatizo ndani ya muda uliokubalika (mfano saa 48), Sales/RM wanamwarifu Admin.
- Admin anatumia *audit\_logs* kuangalia historia kamili ya record husika (nani alifanya nini, lini) kabla ya kufanya marekebisho.
- Marekebisho yoyote ya Admin (mf. kurekebisha amount ya payment, kubadilisha status ya booking kwa mkono) yanaongeza *version* na kuandikwa kwenye *audit\_logs* kama ADMIN\_OVERRIDE, ili kuwe na uwazi wa nini kilifanyika na kwa nini.

## Ngazi ya 4 — Tatizo la kiufundi la system (bug, downtime, upload kushindwa)

- Admin anawasiliana na Platform/Developer Support (nje ya system, mf. email/WhatsApp/ticket ya nje) kwa sababu hili ni tatizo la miundombinu, siyo la data ya kampuni.

## Muhtasari wa muda unaopendekezwa (SLA)

- Password reset: papo hapo (automated)
- Msaada wa ndani ya timu (Ngazi 2): saa 24–48
- Marekebisho ya Admin (Ngazi 3): siku ya kazi 1
- Tatizo la kiufundi (Ngazi 4): kulingana na Platform Support

## Support Tickets — Sasa Imeongezwa Rasmi

Table ya **support\_tickets** imeongezwa kwenye Phase 2 ya database\_schema.pdf, ikifuata muundo uleule wa accommodation\_inquiries lakini kwa masuala ya jumla ya system (login, payment, booking, account, kiufundi):

**id, company\_id, raised\_by, assigned\_to, category, subject, description, related\_entity\_type, related\_entity\_id, priority, status, resolution\_notes, version, resolved\_at, created\_at, updated\_at**

Mfano wa matumizi: mtumiaji "hawezi kuingia" → ticket inaundwa na category=LOGIN\_ISSUE, raised\_by=user mwenyewe au Admin kwa niaba yake, related\_entity\_type='user', related\_entity\_id=id ya user husika. Admin anapoifanyia kazi, anaweka assigned\_to=jina lake, status=IN\_PROGRESS, na akiisha kuitatua anaweka resolution\_notes + status=RESOLVED + resolved\_at. Kila mabadiliko ya ticket yanaingia audit\_logs na version inaongezeka, hivyo kuna historia kamili ya kesi hiyo tangu ilipofunguliwa hadi ilipofungwa.